



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.AT.7

# Properties of Operations

Lesson 6-4 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can use the commutative, associative, and identity properties to rewrite expressions.

**Language Objective:** I can explain my reasoning using the words commutative property, associative property, and identity property.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Commutative Property

Associative Property

Identity Property

Property

Distributive Property



Tap any word to see what it means and a picture.

- 1 The rule that order does not change a sum or product, like  $a + b = b + a$ , is the

- 2 The rule that grouping does not change a sum or product, like  $(a + b) + c = a + (b + c)$ , is the

- 3 The rule that adding 0 or multiplying by 1 keeps a number the same is the

- 4 A rule that is always true in math is a

- 5 The rule that lets you multiply a number across a sum, like  $a(b + c) = a \times b + a \times c$ , is the

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How can you tell the commutative property from the associative property?  
Use  $6 + 9 = 9 + 6$  and  $(3 + 5) + 2 = 3 + (5 + 2)$  to explain?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Commutative Property** — You can change the order and get the same answer.  
*Propiedad conmutativa — Puedes cambiar el orden y obtener la misma respuesta.*

☐ **Associative Property** — You can change the grouping and get the same answer.  
*Propiedad asociativa — Puedes cambiar la agrupación y obtener la misma respuesta.*

☐ **Identity Property** — Adding 0 or multiplying by 1 keeps the same value.  
*Propiedad de identidad — Sumar 0 o multiplicar por 1 mantiene el mismo valor.*

☐ **Property** — A rule that is always true in math.  
*Propiedad — Una regla que siempre es verdadera en matemáticas.*

☐ **Distributive Property** — You can multiply a number by each part inside the parentheses, then add.  
*Propiedad distributiva — Puedes multiplicar un número por cada parte dentro del paréntesis y luego sumar.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The \_\_\_\_ property changes the order, like  $6 + 9 = 9 + 6$ .**

*La propiedad \_\_\_\_ cambia el orden, como  $6 + 9 = 9 + 6$ .*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The commutative property lets you change the order of a sum without changing the total.

**Evidence:** Students name the commutative property and explain that  $3 + 5 + 2$  gives 10 in any order because reordering addends does not change the sum.

**Reasoning:** This evidence connects the definition of commutative property to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The \_\_\_\_ property changes the order, like  $6 + 9 = 9 + 6$ .**

*La propiedad \_\_\_\_ cambia el orden, como  $6 + 9 = 9 + 6$ .*

- **The \_\_\_\_ property changes the grouping because \_\_\_\_.**

*La propiedad \_\_\_\_ cambia la agrupación porque \_\_\_\_.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Commutative Property
2. **Notes 2:** Associative Property
3. **Notes 3:** Identity Property
4. **Notes 4:** Property
5. **Notes 5:** Distributive Property
6. **Watch for this mistake:** *A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume  $15 - 8 = 8 - 15$  (it does not:  $7 \neq -7$  — commutative only works for addition and multiplication), or they see parentheses move and call it "associative" even when the numbers inside just swapped places, like  $(2 + 3) + 4 = (3 + 2) + 4$  (that is commutative — the order inside the parentheses changed, not the grouping). Before you label a property, ask: did only the order change (commutative), did only the grouping change (associative), or was a 0 or 1 involved (identity)?*
7. **Try It 1:** A. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
8. **Try It 2:** A. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*